

## Medical Gas and Vacuum System Failure

|                |                               |                    |                                                                                           |
|----------------|-------------------------------|--------------------|-------------------------------------------------------------------------------------------|
| Policy Number  | POL-EM-055                    | Owner / Department | Facilities / Respiratory Therapy                                                          |
| Effective Date | 2023-09-15                    | Version            | 2.1                                                                                       |
| Last Revised   | 2026-02-24                    | Next Review        | 2028-02-24                                                                                |
| Approved By    | Environment of Care Committee | Applies To         | All clinical staff in areas supplied by medical gases; Facilities and Respiratory Therapy |

### 1. Purpose

To maintain oxygenation and suction for patients during failure, contamination, or depletion of a piped medical gas or vacuum system, and to safely isolate a leaking or involved zone, consistent with NFPA 99 (medical gas and vacuum systems) and The Joint Commission Environment of Care standards.

### 2. Scope

Applies to piped oxygen, medical air, nitrous oxide, and medical-surgical vacuum in all patient-care areas. Clinical staff maintain patient oxygenation with backup cylinders while Facilities and Respiratory Therapy diagnose and restore the system.

### 3. Definitions

- **Zone (shutoff) valve** — a wall valve, usually outside each patient area, that isolates the medical gas supply to that zone.
- **Master alarm / area alarm** — NFPA 99 alarms that annunciate abnormal source or zone pressure to the plant and to staffed areas.
- **E-cylinder** — a portable compressed-gas cylinder used as a backup oxygen source at the bedside or during transport.
- **Medical-surgical vacuum** — the piped suction system used for airway, wound, and gastric suction.

### 4. Procedure — Gas Supply Failure

**IMPORTANT** — Only a qualified person closes a zone valve, and only after confirming which rooms it feeds and that every patient in that zone has an alternate oxygen source (E-cylinder) in place. Closing the wrong valve, or closing one before patients are converted to cylinders, can stop oxygen to ventilated patients. Keep full E-cylinders and regulators immediately available on every unit.

1. Recognize the failure (alarm, loss of flow/pressure, or odor); assess every affected patient's oxygenation.
2. Place patients on backup oxygen (E-cylinders with regulators); provide manual ventilation for ventilated patients if needed and use portable suction.
3. Notify Facilities / the plant operator, the House Supervisor, and Respiratory Therapy immediately; report the gas, area, and symptoms.

4. Isolate only if directed: a qualified person closes the correct zone valve for a leak or fire after confirming all patients are on backup supply.
5. Monitor cylinder levels and arrange resupply; plan transfer of gas-dependent patients if restoration will be prolonged (consider HICS).
6. File an event report per POL-RM-013; log alarm, isolation, and restoration times.

## 5. Severity & Reporting

Medical gas/vacuum failures are classified under POL-RM-013. A loss of oxygen or vacuum to a critical area (ICU, OR, ED, NICU), contaminated gas, or any patient harm is **SEV-1**: notify the Administrator-on-Call, Safety Officer, and Facilities leadership immediately and within 1 hour, and activate HICS for a source-level or facility-wide loss. A localized zone failure managed on backup cylinders without harm is **SEV-2** (within 4 hours). A nuisance alarm or a deficiency found on inspection with no clinical impact is **SEV-3 / Near Miss** within 24 hours.

## 6. Documentation Requirements

- Gas or system affected, zone/area, and time of failure and restoration.
- Alarms that annunciated; whether a zone valve was closed and by whom.
- Patients affected, backup source applied, and any manual support.
- Clinical impact/injury; notifications and HICS status.

## 7. References

- NFPA 99, Health Care Facilities Code — medical gas and vacuum systems, zone valves, and alarms.
- The Joint Commission, Environment of Care — utility systems (medical gas) and inventory of critical components.
- CMS Conditions of Participation / Emergency Preparedness Rule, 42 CFR 482.15.
- Related: POL-RM-013 (severity & reporting), POL-EM-054 (utility / power failure), POL-EM-053 (HICS).